AER1416: Numerical Methods for Uncertainty Quantification
Assignment 1

Zouhair Adam Hamaimou (1004891986)

1 Quadratic Function

P1 Consider the quadratic function in d variables defined below:

$$f(x) = a_0 + \sum_{i=1}^{d} a_i x_i + \gamma \sum_{i=1}^{d} \sum_{j=1}^{d} a_{ij} x_i x_j,$$

where x_i, $i = 1, 2, \ldots, d$ are independent random variables drawn from $\mathcal{N}(0,1)$, γ is a scaling factor, $a_0 = 1$, $a_i = \frac{1}{\sqrt{i}}$, and

$$a_{ij} = \begin{cases} \frac{1}{i}, & \text{if } i = j, \\ \frac{\min(i,j)}{\max(i,j)^2 (i+j)}, & \text{if } i \neq j. \end{cases}$$

(a) Analytical Mean and Variance Derivation

Mean of $f(x)$

$$\mathbb{E}[f(x)] = a_0 + \sum_{i=1}^{d} a_i \underbrace{\mathbb{E}[x_i]}_{=0} + \gamma \sum_{i,j} a_{ij} \underbrace{\mathbb{E}[x_i x_j]}_{=\delta_{ij}} = a_0 + \gamma \sum_{i=1}^{d} a_{ii}.$$

Using $a_0 = 1$ and $a_{ii} = 1/i$,

$$\boxed{\mu_f = \mathbb{E}[f(x)] = 1 + \gamma \sum_{i=1}^{d} \frac{1}{i}.}$$

Variance of $f(x)$

Write $f = a_0 + L + Q$ with $L = \sum_i a_i x_i$ and $Q = \gamma \sum_{i,j} a_{ij} x_i x_j$. Since $\mathrm{Cov}(L, Q) = \gamma \mathbb{E}\Big[\sum_k a_k x_k \sum_{i,j} a_{ij} x_i x_j\Big]$ involves only odd moments $\mathbb{E}[x_k x_i x_j] = 0$ of zero-mean Gaussians, $\mathrm{Cov}(L, Q) = 0$, and

$$\mathrm{Var}[f] = \mathrm{Var}[L] + \mathrm{Var}[Q].$$

Linear term. $\mathrm{Var}[L] = \sum_{i=1}^{d} a_i^2.$

Quadratic term. Expanding $\mathbb{E}[Q^2] = \gamma^2 \sum_{i,j,k,l} a_{ij} a_{kl}\, \mathbb{E}[x_i x_j x_k x_l]$ and applying Isserlis' theorem

$$\mathbb{E}[x_i x_j x_k x_l] = \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$$

gives three pairings:

$$\mathbb{E}[Q^2] = \gamma^2 \left(\sum_i a_{ii} \right)^2 + \gamma^2 \sum_{i,j} a_{ij}^2 + \gamma^2 \sum_{i,j} a_{ij} a_{ji} = (\mathbb{E}[Q])^2 + 2\gamma^2 \sum_{i,j} a_{ij}^2,$$

where symmetry $a_{ij} = a_{ji}$ was used in the third pairing. Hence $\mathrm{Var}[Q] = 2\gamma^2 \sum_{i,j} a_{ij}^2$.

Final result.

$$\boxed{\sigma_f^2 = \mathrm{Var}[f(x)] = \sum_{i=1}^{d} a_i^2 + 2\gamma^2 \sum_{i=1}^{d} \sum_{j=1}^{d} a_{ij}^2.}$$

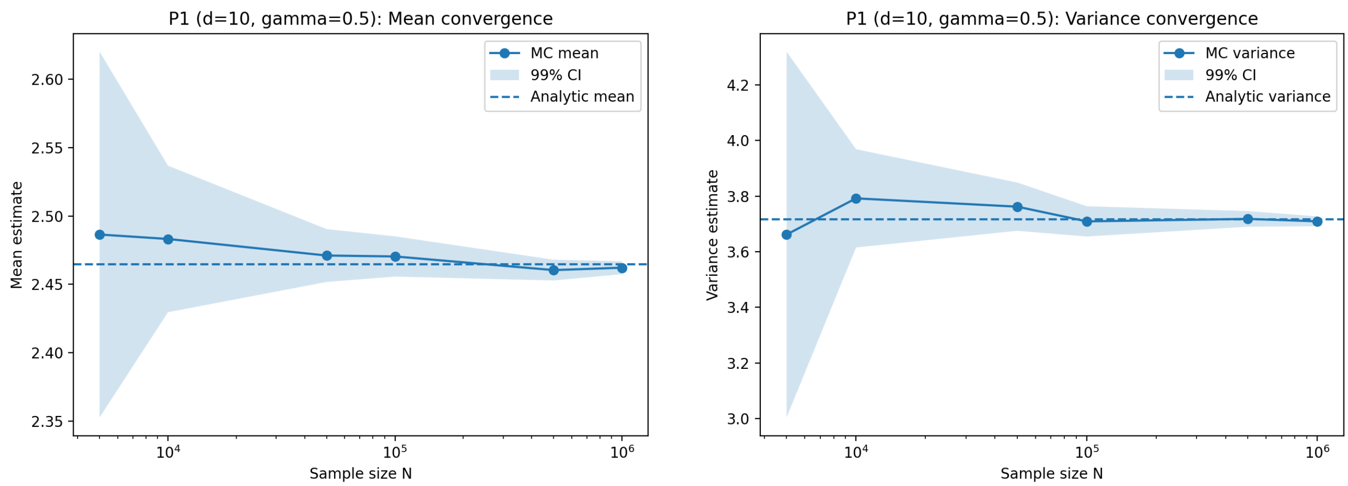
(b) Monte Carlo Approximation ($d = 10$, $\gamma = 0.5$)

We draw i.i.d. samples $x^{(n)} \sim \mathcal{N}(0, I_{10})$ and evaluate $f^{(n)} = f(x^{(n)})$ for $n = 1, \ldots, N$, using the estimators

$$\hat{\mu}_N = \frac{1}{N} \sum_{n=1}^{N} f^{(n)}, \qquad \hat{\sigma}_N^2 = \frac{1}{N-1} \sum_{n=1}^{N} \left(f^{(n)} - \hat{\mu}_N \right)^2.$$

From part (a), the analytical values are $\mu_f = 2.464484$ and $\sigma_f^2 = 3.716711$.

Confidence intervals. For the mean we use $\mathrm{SE}(\hat{\mu}_N) = \hat{\sigma}_N / \sqrt{N}$ and form the 99% CI $\hat{\mu}_N \pm 2.58\,\mathrm{SE}(\hat{\mu}_N)$. For the variance we use the large-N approximation $\mathrm{Var}(\hat{\sigma}_N^2) \approx \frac{1}{N}(\mu_4 - \frac{N-3}{N-1}\sigma^4)$, estimating μ_4 and σ^4 from plug-in sample moments.
Figure 1 shows convergence of both estimators with shrinking 99% CIs. At $N = 5 \times 10^5$ the analytical values lie within the reported intervals; Table 1 summarises the results at this sample size.



(a) Mean convergence (99% CI).

(b) Variance convergence (99% CI).

Figure 1: P1 ($d = 10$, $\gamma = 0.5$): Monte Carlo convergence vs. analytical values.

Quantity	Analytical	Estimate ($N = 5 \times 10^5$)	SE	99% CI
Mean μ_f	2.464484	2.460423	0.002725	[2.453399, 2.467447]
Variance σ_f^2	3.716711	3.718176	0.010003	[3.692372, 3.743980]

Table 1: P1 ($d = 10$, $\gamma = 0.5$): Monte Carlo estimates at $N = 5 \times 10^5$.

The empirical CDF $\widehat{F}_N(z) = \frac{1}{N}\sum_{n=1}^{N} \mathbf{1}\{f^{(n)} \le z\}$ admits a pointwise 99% CI

$$\widehat{F}_N(z) \pm 2.58\sqrt{\widehat{F}_N(z)(1 - \widehat{F}_N(z))/N},$$

which is negligibly narrow for $N = 10^6$; Figure 2 shows the resulting smooth CDF estimate.

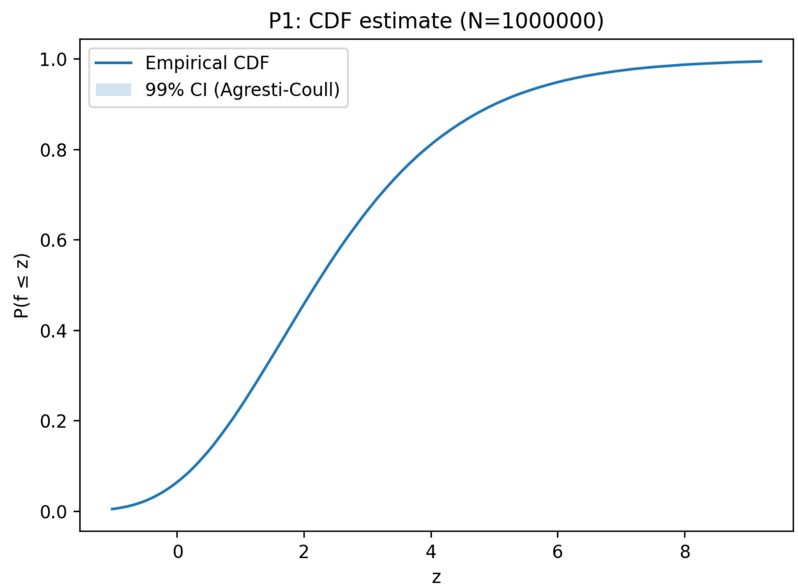


Figure 2: Empirical CDF $\widehat{F}_N(z)$ for P1 with $N = 10^6$.

(c) Central Limit Theorem via Batch Monte Carlo

We partition $N_{\text{tot}} = Bm$ samples into B independent batches of size m. For each batch b, we compute

$$\hat{\mu}^{(b)} = \frac{1}{m}\sum_{n=1}^{m} f_n^{(b)}, \qquad \hat{v}^{(b)} = \frac{1}{m}\sum_{n=1}^{m}\left(f_n^{(b)}\right)^2 - \left(\hat{\mu}^{(b)}\right)^2,$$

then form the batch average $\bar{v} = \frac{1}{B}\sum_b \hat{v}^{(b)}$ with 99% Student-t CI $\bar{v} \pm t_{0.995,B-1}\, s_v/\sqrt{B}$.
For the tuned choice $B = 200$, $m = 10{,}000$ ($N_{\text{tot}} = 2 \times 10^6$),

$$\bar{v} = 3.722995, \qquad 99\%\ \text{CI} = [3.710873,\ 3.735118],$$

which contains $\sigma_f^2 = 3.716711$ (absolute error 6.3×10^{-3}). To verify the CLT, we standardise the batch estimates $Z_b = (\hat{v}^{(b)} - \bar{v})/s_v$; Figure 3 shows their histogram overlaid with the $\mathcal{N}(0,1)$ pdf, confirming an approximately Gaussian distribution.

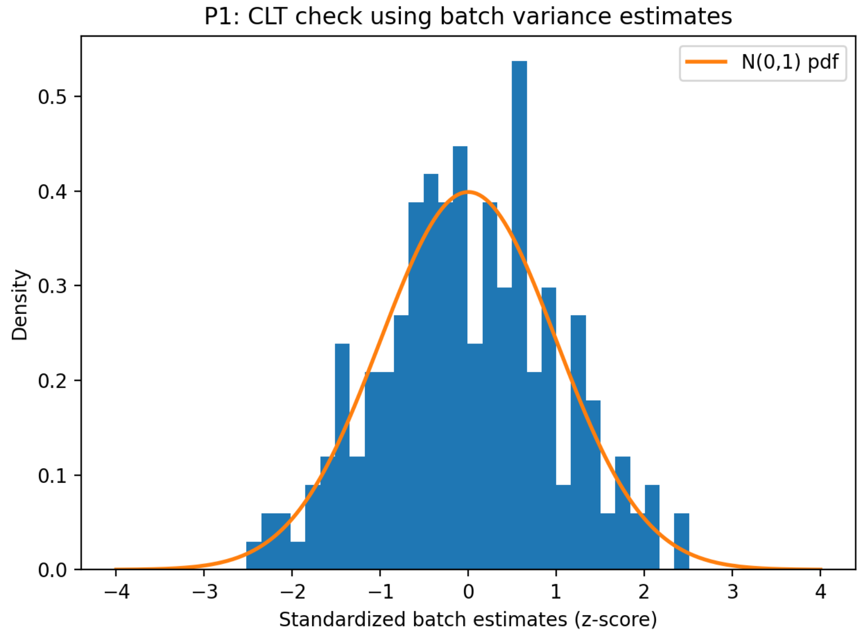


Figure 3: Standardised batch variance estimates Z_b with $\mathcal{N}(0,1)$ overlay ($B = 200$, $m = 10{,}000$).

2 Random Mass–Spring–Damper ODE

2.1 Problem setup

We consider the forced, second-order mass–spring–damper model

$$m\,\ddot{u}(t) + c\,\dot{u}(t) + k\,u(t) = F_0\cos(\omega t), \qquad u(0) = 0,\ \dot{u}(0) = 0, \tag{1}$$

over the time horizon $t \in [0, 10]$ seconds. The parameters are independent random variables:

$$\begin{aligned} &m \sim \mathcal{U}[0.9, 1.1], && c \sim \mathcal{U}[0.8, 1.2], && k \sim \mathcal{U}[10, 12], \\ &\omega \sim \mathcal{U}[0.1, 0.2], && F_0 \sim \mathcal{U}[0.9, 1.1]. \end{aligned} \tag{2}$$

2.2 Monte Carlo procedure

To propagate input uncertainty through the dynamical model, we use a Monte Carlo approach:

1. Draw N i.i.d. samples $\{(m_i, c_i, k_i, \omega_i, F_{0,i})\}_{i=1}^{N}$ from (2).
2. For each sample, solve (1) numerically over $[0, 10]$ using a time-marching ODE solver (here, an adaptive Runge–Kutta scheme).
3. Collect the resulting displacement trajectories $\{u_i(t)\}_{i=1}^{N}$ and estimate the mean and variance of $u(t)$ at each time t.

Let $\hat{\mu}(t)$ and $\hat{\sigma}^2(t)$ denote the Monte Carlo estimators of the mean and variance of $u(t)$. Using the usual sample statistics,

$$\hat{\mu}(t) = \frac{1}{N}\sum_{i=1}^{N} u_i(t), \qquad \hat{\sigma}^2(t) = \frac{1}{N-1}\sum_{i=1}^{N}\big(u_i(t) - \hat{\mu}(t)\big)^2. \tag{3}$$

2.3 Standard errors and 99% confidence intervals

We report 99% confidence intervals using $z_{0.995} \approx 2.58$. The standard error (SE) for the mean estimator is

$$\mathrm{SE}\big(\hat{\mu}(t)\big) = \frac{\sqrt{\hat{\sigma}^2(t)}}{\sqrt{N}}, \qquad \hat{\mu}(t) \ \pm\ 2.58\,\mathrm{SE}\big(\hat{\mu}(t)\big). \tag{4}$$

For the variance estimator, we use the standard large-sample approximation from the prompt,

$$\mathrm{SE}\big(\hat{\sigma}^2(t)\big) \approx \sqrt{\frac{2}{N}}\,\hat{\sigma}^2(t), \qquad \hat{\sigma}^2(t) \ \pm\ 2.58\,\mathrm{SE}\big(\hat{\sigma}^2(t)\big). \tag{5}$$

2.4 Results and discussion

Sample-size study. To determine a suitable Monte Carlo sample size, we first estimated the statistics of $u(10)$ for multiple values of N. Table 2 reports $\hat{\mu}(10)$ and $\hat{\sigma}^2(10)$ along with their standard errors and 99% confidence interval half-widths. As expected, the confidence intervals shrink as N increases, indicating improved statistical accuracy.

Table 2: Monte Carlo estimates for $u(10)$ using multiple sample sizes. SE denotes standard error, and the reported 99% CI half-width equals $2.58 \times$ SE.

Samples	$\hat{\mu}(10)$	SE($\hat{\mu}$)	±99% CI($\hat{\mu}$)	$\hat{\sigma}^2(10)$	SE($\hat{\sigma}^2$)	±99% CI($\hat{\sigma}^2$)
1,000	0.007810	0.000805	0.002076	0.000648	0.000029	0.000075
5,000	0.007559	0.000357	0.000920	0.000636	0.000013	0.000033
10,000	0.007623	0.000252	0.000651	0.000637	0.000009	0.000023
50,000	0.007231	0.000114	0.000295	0.000652	0.000004	0.000011
200,000	0.007129	0.000057	0.000147	0.000649	0.000002	0.000005

Full time-history statistics. Using a larger run ($N = 200{,}000$), we estimated $\hat{\mu}(t)$ and $\hat{\sigma}^2(t)$ over $t \in [0, 10]$ on a uniform time grid. Figure 4 shows the resulting mean and variance trajectories with 99% confidence bands computed using (4)–(5). The mean response exhibits a transient peak followed by damped oscillations, consistent with a forced, damped oscillator. The variance starts at zero (since all realizations satisfy $u(0) = 0$) and grows over time, reaching its largest values near the end of the horizon, reflecting the cumulative effect of parameter uncertainty through the dynamics.

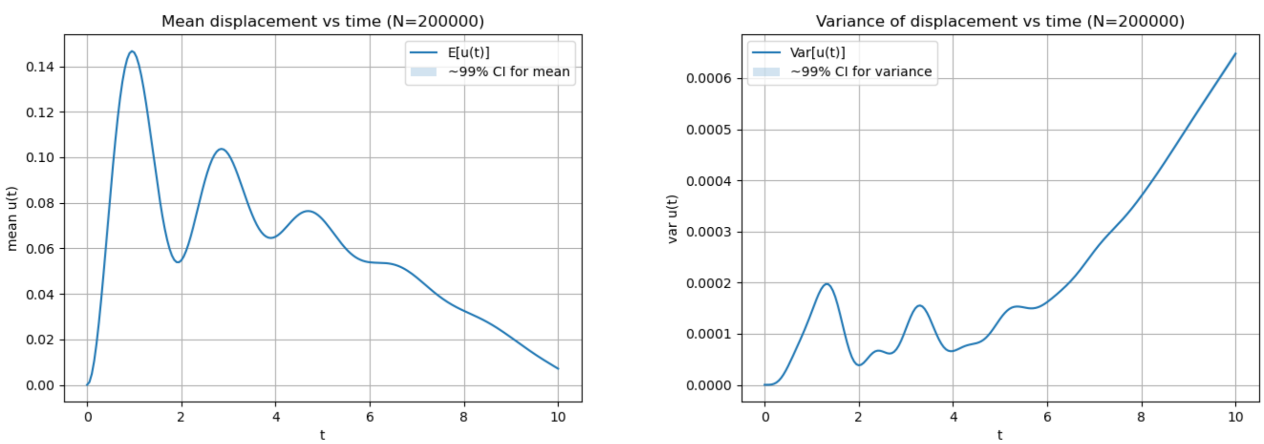


Figure 4: Estimated mean $\hat{\mu}(t)$ (left) and variance $\hat{\sigma}^2(t)$ (right) of the displacement over $t \in [0, 10]$, including 99% confidence bands (large-sample approximations).

3 Random Linear System

3.1 Problem setup

We consider the random linear system arising from the stochastic diffusion model in Problem 3:

$$A(\boldsymbol{\theta})\, u(\boldsymbol{\theta}) = b, \qquad A(\boldsymbol{\theta}) = A_0 + \sum_{i=1}^{3} \theta_i A_i, \tag{6}$$

where $A_0, A_1, A_2, A_3 \in \mathbb{R}^{n\times n}$, $b \in \mathbb{R}^n$, and $n = 143$. The random inputs are assumed independent and uniformly distributed as

$$\theta_i \sim \mathcal{U}[-3, 3], \qquad i = 1, 2, 3, \tag{7}$$

with $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3)^\top$. Our objective is to approximate the componentwise mean and variance of the solution vector $u(\boldsymbol{\theta}) \in \mathbb{R}^{143}$.

3.2 Monte Carlo procedure

We use a crude Monte Carlo approach:

1. Draw N i.i.d. samples $\{\boldsymbol{\theta}^{(k)}\}_{k=1}^{N}$ from (7).
2. For each sample, assemble $A(\boldsymbol{\theta}^{(k)})$ using (6) and solve

$$A(\boldsymbol{\theta}^{(k)})\, u^{(k)} = b. \tag{8}$$

3. Estimate the mean and variance of each component u_j (for $j = 1, \dots, n$) using sample statistics.

Let $\hat{\mu}_j$ and $\hat{\sigma}_j^2$ denote the Monte Carlo estimators of the mean and variance of the jth component of u. With $\{u_j^{(k)}\}_{k=1}^{N}$ denoting the jth component of the kth solution,

$$\hat{\mu}_j = \frac{1}{N}\sum_{k=1}^{N} u_j^{(k)}, \qquad \hat{\sigma}_j^2 = \frac{1}{N-1}\sum_{k=1}^{N}\big(u_j^{(k)} - \hat{\mu}_j\big)^2. \tag{9}$$

We report 99% confidence intervals using $z_{0.995} \approx 2.58$. For each component j,

$$\hat{\mu}_j \pm 2.58\sqrt{\frac{\hat{\sigma}_j^2}{N}}, \qquad \hat{\sigma}_j^2 \pm 2.58\sqrt{\frac{2}{N}}\,\hat{\sigma}_j^2. \tag{10}$$

3.3 Results and discussion

To assess convergence with respect to the Monte Carlo sample size, we report results for a representative component (here, u_1). Table 3 shows $\hat{\mu}_1$ and $\hat{\sigma}_1^2$ along with standard errors and 99% confidence interval half-widths. As N increases, the estimates stabilize and the confidence intervals shrink, consistent with the expected $O(N^{-1/2})$ Monte Carlo accuracy.
Using a larger run (N = 200,000), we estimate the full mean vector $\hat{\boldsymbol{\mu}}$ and variance vector $\hat{\boldsymbol{\sigma}}^2$ across all spatial degrees of freedom. Figure 5 visualizes the spatial distribution of the mean and variance in both 3D surface and contour form. Because the PDE uses Dirichlet boundary conditions (solution constrained to be near zero at the domain boundary), both the mean and variance decrease toward the edges. The mean field exhibits a pronounced peak near the center of the domain, while the variance is comparatively small (on the order of 10^{-6}), indicating limited sensitivity of the solution to the random perturbations over the specified input range.

Table 3: Monte Carlo estimates of the first component $u_1(\boldsymbol{\theta})$ for different sample sizes N. SE denotes standard error, and the reported 99% CI half-width equals $2.58 \times$ SE.

N	$\hat{\mu}_1$	SE($\hat{\mu}_1$)	±99% CI($\hat{\mu}_1$)	$\hat{\sigma}_1^2$	SE($\hat{\sigma}_1^2$)	±99% CI($\hat{\sigma}_1^2$)
1,000	$6.40227e{-}02$	$2.699e{-}04$	$6.963e{-}04$	$7.28373e{-}05$	$3.257e{-}06$	$8.404e{-}06$
5,000	$6.35569e{-}02$	$1.191e{-}04$	$3.072e{-}04$	$7.08757e{-}05$	$1.418e{-}06$	$3.657e{-}06$
10,000	$6.36429e{-}02$	$8.497e{-}05$	$2.192e{-}04$	$7.21933e{-}05$	$1.021e{-}06$	$2.634e{-}06$
50,000	$6.36593e{-}02$	$3.800e{-}05$	$9.805e{-}05$	$7.22178e{-}05$	$4.567e{-}07$	$1.178e{-}06$
200,000	$6.36061e{-}02$	$1.899e{-}05$	$4.898e{-}05$	$7.20872e{-}05$	$2.280e{-}07$	$5.881e{-}07$

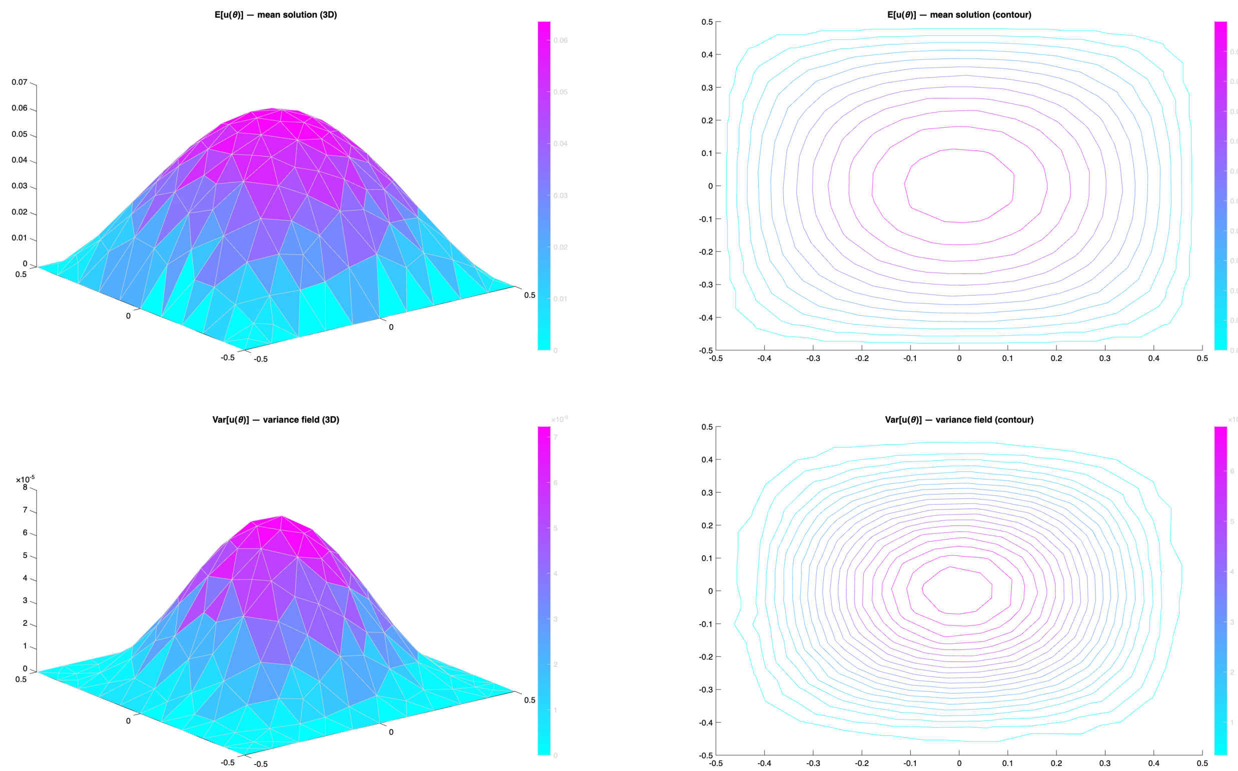


Figure 5: Monte Carlo results for the large run ($N = 200{,}000$). Top row: mean field $\hat{\boldsymbol{\mu}}$ (3D surface and contour). Bottom row: variance field $\hat{\boldsymbol{\sigma}}^2$ (3D surface and contour).

Conclusion. Monte Carlo simulation provides consistent estimates of the mean and variance of the solution field $u(\boldsymbol{\theta})$. The convergence study confirms narrowing 99% confidence intervals as N increases, and the large-N field plots reveal a mean response peaked near the center with small spatial variance, consistent with the expected behavior of a diffusion problem with Dirichlet boundaries.

4 First-Order (Linear) Approximation

Linearization Method

For a quantity of interest $\phi(\alpha)$ depending on independent random parameters $\alpha = (\alpha_1, \ldots, \alpha_p)$ with means $\bar{\alpha} = (\bar{\alpha}_1, \ldots, \bar{\alpha}_p)$, the first-order Taylor expansion about $\bar{\alpha}$ gives

$$\tilde{\phi}(\alpha) = \phi(\bar{\alpha}) + \sum_{i=1}^{p} \frac{\partial \phi}{\partial \alpha_i}\bigg|_{\bar{\alpha}} (\alpha_i - \bar{\alpha}_i).$$

Since $\tilde{\phi}$ is a linear (affine) function of independent zero-mean deviations, the mean and variance follow immediately:

$$\mathbb{E}[\tilde{\phi}] = \phi(\bar{\alpha}), \qquad \mathrm{Var}[\tilde{\phi}] = \sum_{i=1}^{p} \left(\frac{\partial \phi}{\partial \alpha_i}\bigg|_{\bar{\alpha}} \right)^2 \mathrm{Var}(\alpha_i).$$

P1: Quadratic Function

Setup. We have $f(x) = a_0 + \sum_{i=1}^{d} a_i x_i + \gamma \sum_{i,j} a_{ij} x_i x_j$ with $d = 10$, $\gamma = 0.5$, $x_i \sim \mathcal{N}(0,1)$, and expansion point $z = 0$ (the mean of x).
First-order expansion. Since $f(0) = a_0 = 1$ and $\frac{\partial f}{\partial x_i}\big|_{x=0} = a_i = \frac{1}{\sqrt{i}}$, the linear approximation is

$$\tilde{f}(x) = 1 + \sum_{i=1}^{10} \frac{1}{\sqrt{i}} x_i.$$

Approximate mean and variance. Since $\mathbb{E}[x_i] = 0$ and $\mathrm{Var}(x_i) = 1$,

$$\mathbb{E}[\tilde{f}] = 1, \qquad \mathrm{Var}[\tilde{f}] = \sum_{i=1}^{10} \frac{1}{i} \approx 2.928968.$$

Comparison to analytical values. Table 4 compares the linear approximation to the analytical results from Section 1.

Quantity	Analytical	Linear Approx	Absolute Error
Mean μ_f	2.464484	1.000000	1.464484
Variance σ_f^2	3.716711	2.928968	0.787743

Table 4: P1 ($d = 10$, $\gamma = 0.5$): first-order approximation vs. analytical values.

The linear approximation substantially underestimates both the mean and variance. The mean error of 1.464 arises because the entire quadratic contribution $\gamma \sum_{i=1}^{d} a_{ii} = 0.5 \sum_{i=1}^{10} \frac{1}{i} \approx 1.464$ is discarded when the expansion is truncated after the linear term. Similarly, the variance error of 0.788 reflects the missing quadratic contribution $2\gamma^2 \sum_{i,j} a_{ij}^2$. The approximation degrades as γ increases, since the function becomes increasingly nonlinear.

P2: Random Mass-Spring-Damper ODE

Setup. The displacement $u(t)$ satisfies

$$m\ddot{u} + c\dot{u} + ku = F_0 \cos(\omega t), \qquad u(0) = \dot{u}(0) = 0,$$

with independent uniform parameters $m \sim \mathcal{U}[0.9, 1.1]$, $c \sim \mathcal{U}[0.8, 1.2]$, $k \sim \mathcal{U}[10, 12]$, $\omega \sim \mathcal{U}[0.1, 0.2]$, $F_0 \sim \mathcal{U}[0.9, 1.1]$. The nominal values are the means: $\bar{m} = 1$, $\bar{c} = 1$, $\bar{k} = 11$, $\bar{\omega} = 0.15$, $\bar{F}_0 = 1$. We evaluate the approximation at $t = 10$.
Partial derivatives. Each partial derivative of $u(10)$ is approximated by a forward finite difference with step $\varepsilon = 10^{-4}$:

$$\frac{\partial u}{\partial \alpha}\bigg|_{\bar{\alpha}}(10) \approx \frac{u(10;\, \bar{\alpha} + \varepsilon\, e_\alpha) - u(10;\, \bar{\alpha})}{\varepsilon}.$$

Table 5 lists the computed derivatives and their contribution to the variance.

Param. α	Range	$\mathrm{Var}(\alpha)$	$\partial u/\partial\alpha\big\vert_{t=10}$	$(\partial u/\partial\alpha)^2\mathrm{Var}(\alpha)$
m	[0.9, 1.1]	3.333×10^{-3}	-0.010339	3.566×10^{-7}
c	[0.8, 1.2]	1.333×10^{-2}	$+0.001769$	4.174×10^{-8}
k	[10, 12]	3.333×10^{-1}	$+0.000100$	3.333×10^{-9}
F_0	[0.9, 1.1]	3.333×10^{-3}	$+0.007470$	1.860×10^{-7}
ω	[0.1, 0.2]	8.333×10^{-4}	-0.899172	6.735×10^{-4}
			Total $\mathrm{Var}[\tilde{u}(10)]$	6.736×10^{-4}

Table 5: P2: finite-difference partial derivatives and their contributions to $\mathrm{Var}[\tilde{u}(10)]$.

Approximate mean and variance. Since each parameter has zero mean deviation,

$$\mathbb{E}[\tilde{u}(10)] = u_0(10) = 0.007470, \qquad \mathrm{Var}[\tilde{u}(10)] = 6.736\times10^{-4}.$$

Comparison to Monte Carlo (Q2). Table 6 compares the linear approximation to the largest-N Monte Carlo estimates from Section 2 ($N = 200{,}000$).

Quantity	MC ($N = 200{,}000$)	Linear Approx	Relative Error
$\mathbb{E}[u(10)]$	0.007129	0.007470	4.8%
$\mathrm{Var}[u(10)]$	6.490×10^{-4}	6.736×10^{-4}	3.8%

Table 6: P2: first-order approximation vs. Monte Carlo at $t = 10$.

The linear approximation agrees closely with the Monte Carlo estimates, with relative errors below 5% for both mean and variance. The dominant sensitivity is to ω, which accounts for over 99% of $\mathrm{Var}[\tilde{u}(10)]$. This reflects the high sensitivity of the forced oscillator's steady-state amplitude to the driving frequency relative to the natural frequency $\omega_n = \sqrt{k/m} \approx \sqrt{11} \approx 3.32\,\mathrm{rad/s}$. The remaining parameters (m, c, k, F_0) contribute negligibly by comparison. The small errors in mean and variance confirm that the system's response at $t = 10$ is well approximated as a linear function of the input parameters over their respective ranges.

P3: Random Linear System

Setup. We consider the random linear system

$$A(\boldsymbol{\theta})\,u(\boldsymbol{\theta}) = b, \qquad A(\boldsymbol{\theta}) = A_0 + \sum_{i=1}^{3}\theta_i A_i, \qquad \theta_i \sim \mathcal{U}[-3,+3],$$

where $A_0, A_1, A_2, A_3 \in \mathbb{R}^{143\times143}$ and $b \in \mathbb{R}^{143}$. We seek a first-order approximation of the solution vector $u(\boldsymbol{\theta})$.

First-order expansion. We expand about $\boldsymbol{\theta} = 0$. At this point $A(0) = A_0$, so the nominal solution satisfies

$$A_0\,u(0) = b \implies u(0) = A_0^{-1}b.$$

Differentiating $A(\boldsymbol{\theta})\,u(\boldsymbol{\theta}) = b$ with respect to θ_i and evaluating at $\boldsymbol{\theta} = 0$ gives

$$A_i\,u(0) + A_0\,\frac{\partial u}{\partial\theta_i}\bigg|_{\boldsymbol{\theta}=0} = 0, \qquad \text{so} \qquad \frac{\partial u}{\partial\theta_i}\bigg|_{\boldsymbol{\theta}=0} = -A_0^{-1}A_i\,u(0).$$

Defining the sensitivity vectors $\beta_i = -A_0^{-1}A_i\,u(0) \in \mathbb{R}^{143}$, the linear approximation is

$$\tilde{u}(\boldsymbol{\theta}) = u(0) + \sum_{i=1}^{3}\beta_i\,\theta_i.$$

Approximate mean and variance. Since $\theta_i \sim \mathcal{U}[-3, +3]$ are independent with

$$\mathbb{E}[\theta_i] = 0, \qquad \mathrm{Var}(\theta_i) = \frac{(3-(-3))^2}{12} = \frac{36}{12} = 3,$$

the mean and variance of the linear approximation are

$$\mathbb{E}[\tilde{u}(\boldsymbol{\theta})] = u(0), \qquad \mathrm{Var}[\tilde{u}(\boldsymbol{\theta})] = \sum_{i=1}^{3} \big(\beta_i \beta_i^T\big) \mathrm{Var}(\theta_i) = 3 \sum_{i=1}^{3} \beta_i \beta_i^T.$$

The result is a 143×143 matrix whose diagonal entries give the componentwise variances.
Comparison to Monte Carlo (Q3). Table 7 focuses on the first component of $u(\boldsymbol{\theta})$ and compares the linear approximation to the largest-N Monte Carlo estimate from Section 3 ($N = 200{,}000$).

Quantity	MC ($N = 200{,}000$)	Linear Approx	Relative Error
$\mathbb{E}[u_1(\boldsymbol{\theta})]$	6.36061×10^{-2}	6.24029×10^{-2}	1.89%
$\mathrm{Var}[u_1(\boldsymbol{\theta})]$	7.20872×10^{-5}	6.61750×10^{-5}	8.20%

Table 7: P3: first-order approximation vs. Monte Carlo for the first solution component u_1.

The linear approximation recovers the mean with a relative error below 2% and the variance with a relative error of roughly 8%. Because $A(\boldsymbol{\theta})$ depends linearly on $\boldsymbol{\theta}$, the system is well-suited to a first-order approach, and the small discrepancies reflect the mild nonlinearity introduced through the matrix inverse $(A_0 + \sum_i \theta_i A_i)^{-1}$ over the wide input range $[-3, +3]$. The approximation is notably more accurate for P3 than for P1, where the explicit quadratic term caused large errors, confirming that linearization performs best when the underlying map is close to linear.

Overall Conclusions

The accuracy of the first-order approximation depends strongly on the degree of nonlinearity of the forward map over the parameter ranges:

- **P1**: The linear expansion discards the quadratic term entirely, leading to large errors in both mean ($\Delta\mu \approx 1.46$) and variance ($\Delta\sigma^2 \approx 0.79$). Linearization is a poor substitute for the full analytical result when γ is not small.
- **P2**: Expanding about the nominal parameter values yields mean and variance estimates within 5% of the Monte Carlo values at $t = 10$. The forcing frequency ω dominates the variance, accounting for over 99% of $\mathrm{Var}[\tilde{u}(10)]$.
- **P3**: Because $A(\boldsymbol{\theta})$ is linear in $\boldsymbol{\theta}$, the first-order expansion is the most accurate of the three cases, with mean and variance errors below 2% and 9% respectively, despite the broad input range $[-3, +3]$.

5 Control Variate Method for P1

Setup

We apply the control variate (CV) technique to estimate $\mathbb{E}[f(x)]$ for P1 with $d = 10$ and $x_i \sim \mathcal{N}(0, 1)$, comparing standard MC against CV MC for $\gamma = 0.5$ and $\gamma = 0.1$.

Control Variate

We choose the linear part of f as the control variate,

$$g(x) = a_0 + \sum_{i=1}^{d} a_i x_i = 1 + \sum_{i=1}^{d} \frac{1}{\sqrt{i}}\, x_i,$$

with known mean $\mu_g = \mathbb{E}[g(x)] = 1$ since $\mathbb{E}[x_i] = 0$.

CV Estimator

Given $\lambda^* = \mathrm{Cov}[f, g]/\mathrm{Var}[g]$, estimated from the same sample, the CV estimator of $\mathbb{E}[f]$ is

$$\widehat{I}_{\mathrm{CV}} = \frac{1}{N} \sum_{i=1}^{N} \Big[f(x^{(i)}) - \widehat{\lambda}^* \big(g(x^{(i)}) - \mu_g \big) \Big] .$$

This reduces variance whenever f and g are correlated. At $N = 200{,}000$ the estimated optimal coefficient converges to $\widehat{\lambda}^* \approx 1$ for both values of γ, consistent with the fact that g is the linear part of f.

Results

Case $\gamma = 0.5$. The analytical mean is $\mu_f = 2.464484$.

N	$\hat{\mu}_{\mathrm{MC}} \pm \mathrm{CI}$	$\hat{\sigma}^2_{\mathrm{MC}} \pm \mathrm{CI}$	$\hat{\mu}_{\mathrm{CV}} \pm \mathrm{CI}$	$\hat{\sigma}^2_{\mathrm{CV}} \pm \mathrm{CI}$	$\frac{\mathrm{Var(CV)}}{\mathrm{Var(MC)}}$
1,000	2.47643 ± 0.15735	$3.7196 \pm 4.292 \times 10^{-1}$	2.47859 ± 0.07892	$9.357 \times 10^{-1} \pm 1.080 \times 10^{-1}$	0.252
5,000	2.49350 ± 0.07236	$3.9333 \pm 2.030 \times 10^{-1}$	2.49333 ± 0.03378	$8.573 \times 10^{-1} \pm 4.424 \times 10^{-2}$	0.218
10,000	2.48744 ± 0.05061	$3.8473 \pm 1.404 \times 10^{-1}$	2.46930 ± 0.02291	$7.887 \times 10^{-1} \pm 2.878 \times 10^{-2}$	0.205
50,000	2.45802 ± 0.02221	$3.7053 \pm 6.046 \times 10^{-2}$	2.46596 ± 0.01030	$7.971 \times 10^{-1} \pm 1.301 \times 10^{-2}$	0.215
200,000	2.46290 ± 0.01109	$3.6965 \pm 3.016 \times 10^{-2}$	2.46473 ± 0.00511	$7.854 \times 10^{-1} \pm 6.407 \times 10^{-3}$	0.213

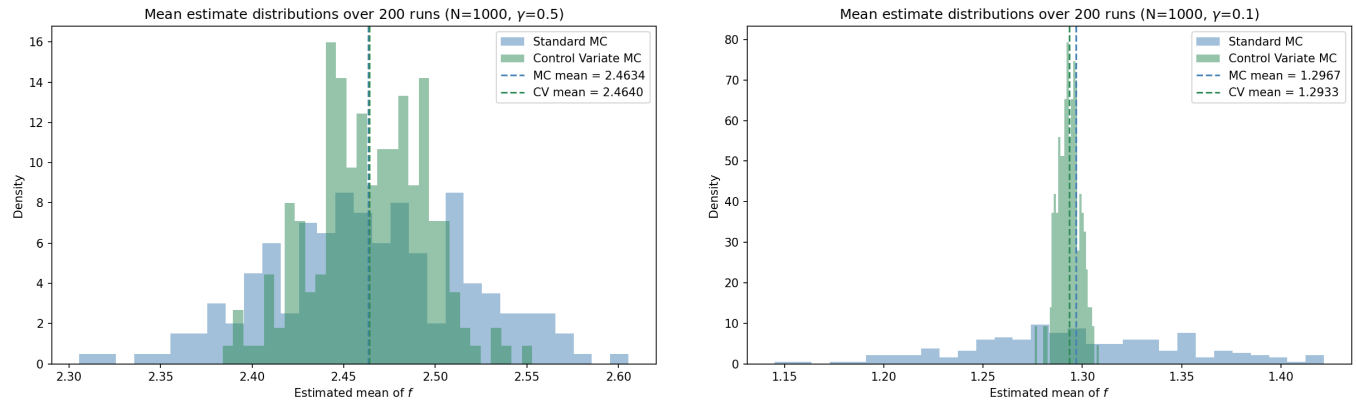
Table 8: Q5 results for $\gamma = 0.5$, $d = 10$. CI denotes the 99% confidence interval half-width ($z_{0.995} = 2.58$).

Case $\gamma = 0.1$. The analytical mean is $\mu_f = 1.292897$.

N	$\hat{\mu}_{\mathrm{MC}} \pm \mathrm{CI}$	$\hat{\sigma}^2_{\mathrm{MC}} \pm \mathrm{CI}$	$\hat{\mu}_{\mathrm{CV}} \pm \mathrm{CI}$	$\hat{\sigma}^2_{\mathrm{CV}} \pm \mathrm{CI}$	$\frac{\mathrm{Var(CV)}}{\mathrm{Var(MC)}}$
1,000	1.37796 ± 0.14213	$3.0347 \pm 3.501 \times 10^{-1}$	1.29132 ± 0.01416	$3.014 \times 10^{-2} \pm 3.478 \times 10^{-3}$	0.010
5,000	1.30432 ± 0.06337	$3.0167 \pm 1.557 \times 10^{-1}$	1.29634 ± 0.00661	$3.287 \times 10^{-2} \pm 1.696 \times 10^{-3}$	0.011
10,000	1.30524 ± 0.04467	$2.9983 \pm 1.094 \times 10^{-1}$	1.29186 ± 0.00455	$3.111 \times 10^{-2} \pm 1.135 \times 10^{-3}$	0.010
50,000	1.28889 ± 0.01982	$2.9503 \pm 4.814 \times 10^{-2}$	1.29191 ± 0.00204	$3.113 \times 10^{-2} \pm 5.080 \times 10^{-4}$	0.011
200,000	1.29280 ± 0.00992	$2.9580 \pm 2.413 \times 10^{-2}$	1.29333 ± 0.00103	$3.164 \times 10^{-2} \pm 2.581 \times 10^{-4}$	0.011

Table 9: Q5 results for $\gamma = 0.1$, $d = 10$.

Figure 6 shows the distribution of mean estimates across $M = 200$ independent runs at $N = 1{,}000$.



(a) $\gamma = 0.5$: moderate variance reduction ($\approx \times 0.21$).

(b) $\gamma = 0.1$: strong variance reduction ($\approx \times 0.01$).

Figure 6: Distribution of mean estimates (standard MC vs. CV) over $M = 200$ runs at $N = 1{,}000$.

Discussion

For $\gamma = 0.5$, f has a substantial quadratic component, so f and g are only moderately correlated. The CV reduces the estimator variance by a factor of approximately 0.21, a 79% reduction. The CI half-width for the mean shrinks from ± 0.011 (standard MC) to ± 0.005 (CV) at $N = 200{,}000$.

For $\gamma = 0.1$, f is nearly linear, making f and g highly correlated. The variance ratio drops to approximately 0.011, a 99% reduction. The CV CI half-width at $N = 200{,}000$ is ± 0.001, roughly ten times tighter than standard MC at the same sample size.

In both cases $\widehat{\lambda}^* \approx 1$, confirming that the CV effectively subtracts the entire linear part of f, leaving only the residual quadratic fluctuations. The effectiveness of the method therefore scales directly with how closely f resembles its linear control variate g, i.e. how small γ is.